

Andrew McAdams

DAT 350

MP1: Getting Data-Driven Answers from Surveys

2/5/26

[Survey Link](#)

[GitHub Link](#)

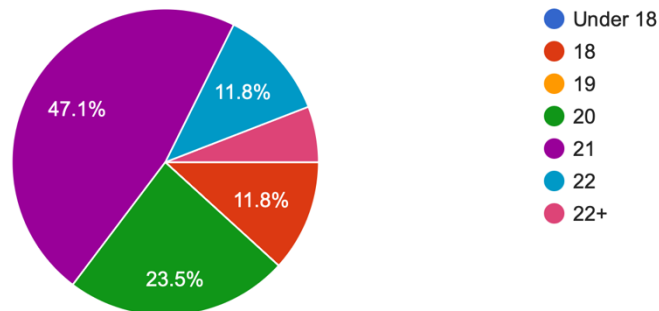
Story:

Intro:

The purpose of the project was to design a survey and learn how to use Python to analyze data from the survey results and then draw insights about the data. The survey focuses on respondents' demographics, their preferences about Sports Viewing Habits, and behaviors showing potential motivations behind viewing habits. The main goal was to ask what sport is watched most often. After that, the idea is to see what factors influence each respondent's most watched sport, like age, academic year, student athlete status, and when they watch, where they watch, and why they watch the sport they view most often.

Who:

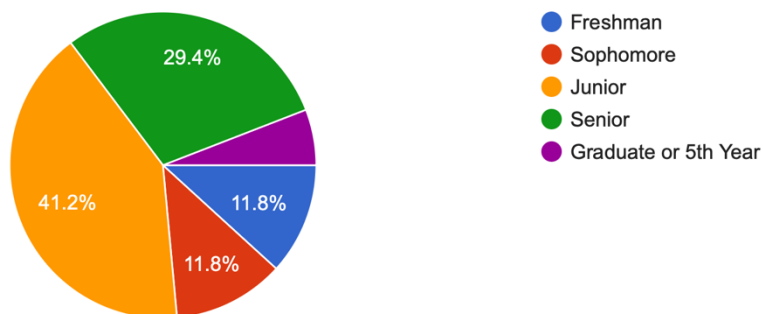
First, to understand the demographics of the survey, respondents were asked about their age, academic status, major, and student athlete status. Percentages were calculated for each category. The first question asked was age, and as shown, the most common age was 22 years old. Both the graph and Python results are pictured below.



```
age
21    47.058824
20    23.529412
18    11.764706
22    11.764706
22+    5.882353
```

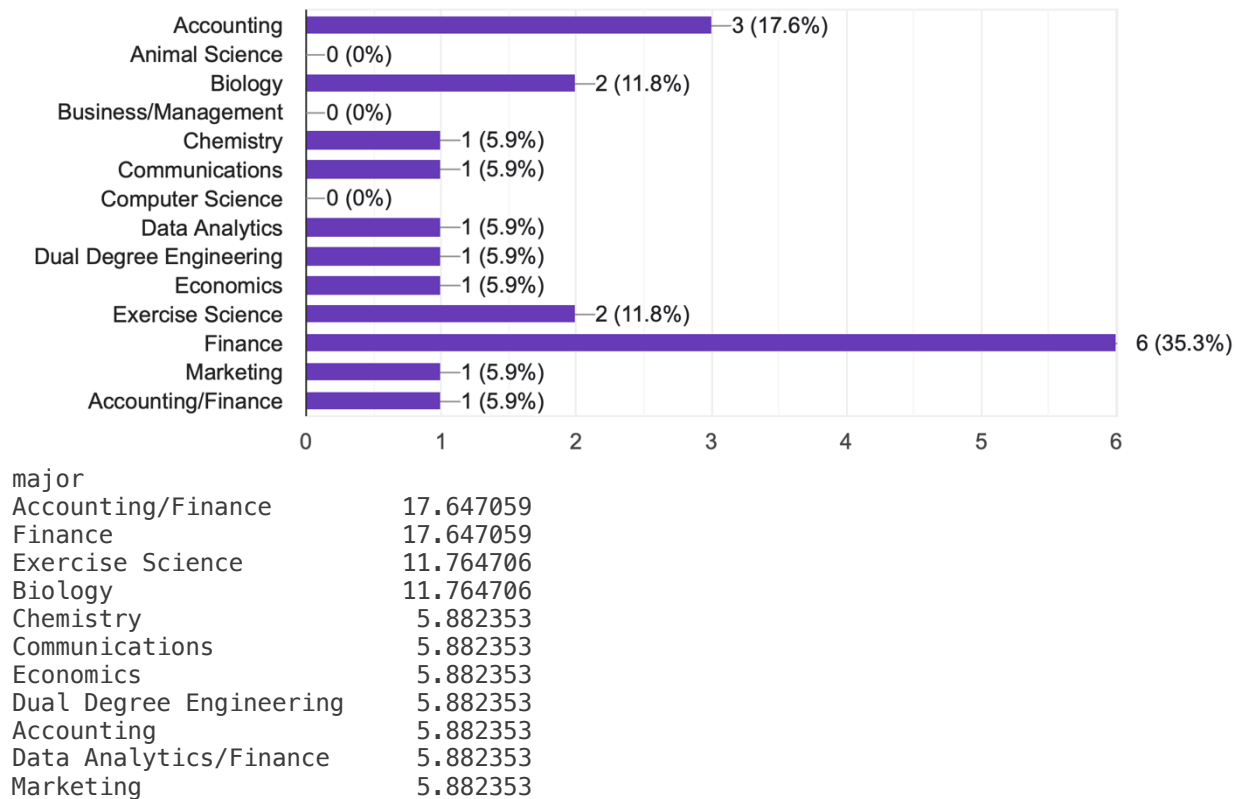
The next question asked was academic year, and as shown, the most common year was a Junior.

Both the graph and Python results are pictured below.

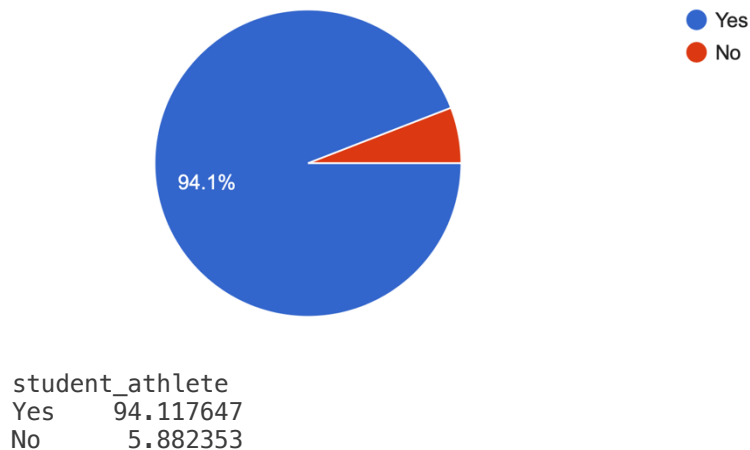


```
academic_year
Junior    41.176471
Senior    29.411765
Freshman  11.764706
Sophomore 11.764706
Graduate or 5th Year  5.882353
```

The next question asked was major, and as shown, the most common major was Finance. Both the graph and Python results are pictured below.

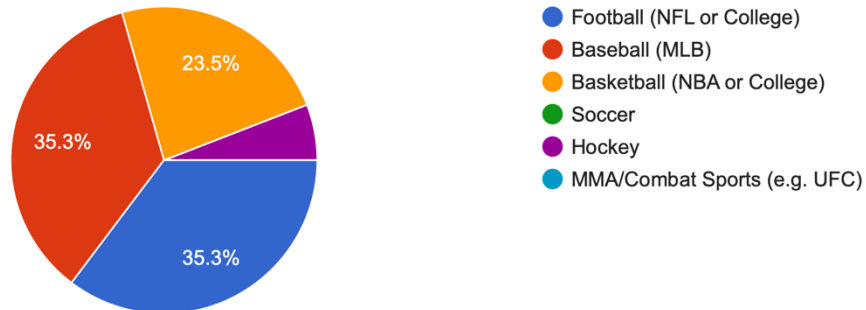


The final demographic question asked was student-athlete status, and as shown, all but one respondent is a student-athlete. Both the graph and Python results are pictured below.



What:

Next, respondents were asked which sport they watch most often. Percentages were calculated for each category to determine the most commonly viewed sport in our sample. We see the graph of all the options along with the Python results.

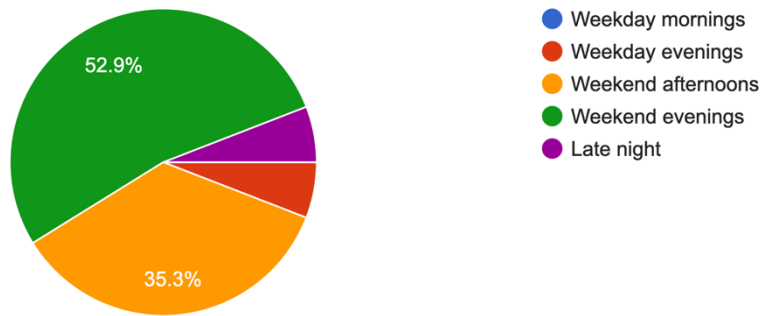


Baseball (MLB)	35.294118
Football (NFL or College)	35.294118
Basketball (NBA or College)	23.529412
Hockey	5.882353

As you can see, Baseball and Football are tied for the most commonly viewed sports. Basketball follows fairly closely behind those two, with a steep drop off at fourth for Hockey.

When:

Moving forward, the survey then checks to see when respondents most often watch sports. Again, percentages were calculated to determine the most popular times among the sample, and included below is the graph and Python results.

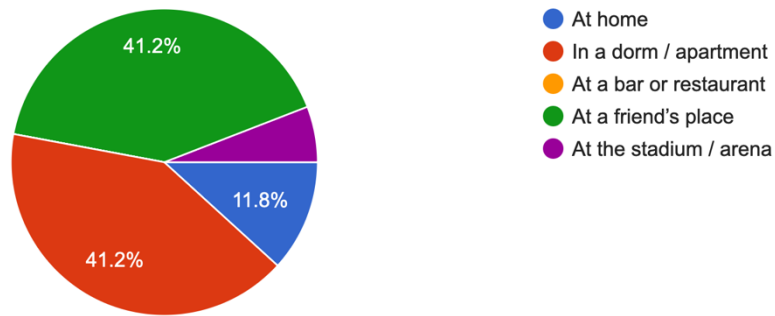


```
when
Weekend evenings    52.941176
Weekend afternoons  35.294118
Late night          5.882353
Weekday evenings    5.882353
```

Overall, we see that the most common time for viewing sports is unsurprisingly on weekend evenings, followed by weekend afternoons. After these top two, we see a very steep drop off to late-night and weekday evening viewing.

Where:

Furthermore, the survey then checks to see where respondents most often watch sports. As previously mentioned, percentages were calculated to determine the most popular locations among the sample and included below is the graph and Python results.



```
where
In a dorm / apartment      41.176471
At a friend's place        41.176471
At home                    11.764706
At the stadium / arena     5.882353
```

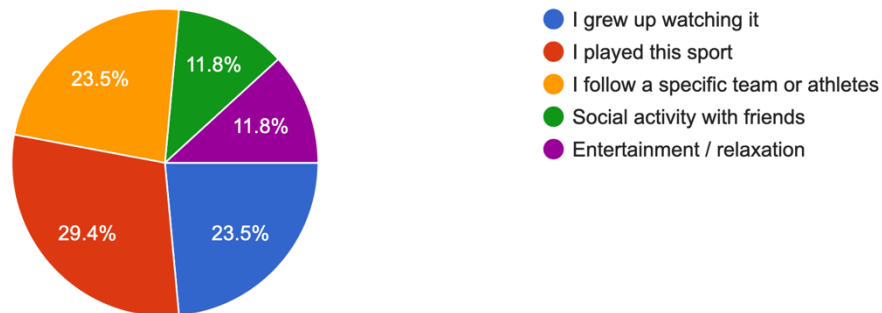
Here, we see that the most common spot for viewing sports is tied for either in the dorm/apartment or at a friend's place. Next, we see people watching at home, and then last is at the actual event. These aren't too surprising results since many people watch sports to engage with others. Also, it is difficult, time-consuming, and costly to attend lots of sports events in person, so this adds to it being last.

Why:

Finally, there are several questions assessing why the respondents make such viewing choices. The survey asks why they watch the sport they chose, why they watch the most at a particular time, and why they watch in a particular location. The results and graphs are pictured below.

Why do you watch this sport most often?

17 responses

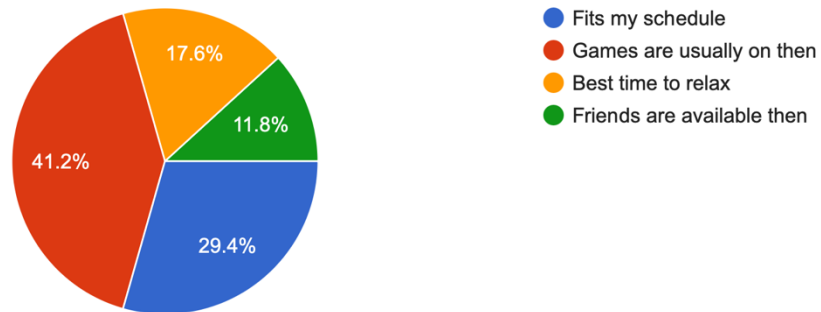


why	
I played this sport	29.411765
I follow a specific team or athletes	23.529412
I grew up watching it	23.529412
Social activity with friends	11.764706
Entertainment/relaxation	11.764706

We see here that most people are motivated to watch their most commonly viewed sport by the fact that they either played the sport, follow a specific team, or grew up watching. All these intuitively make sense and are reflected by the intense fandom and dedication to teams across the country.

Why do you usually watch sports at that time?

17 responses

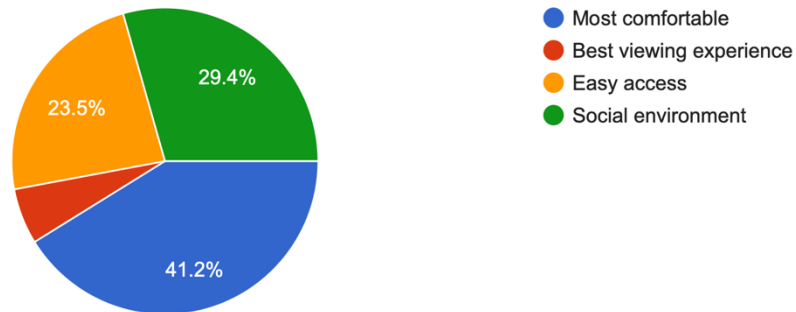


```
why_time
Games are usually on then    41.176471
Fits my schedule             29.411765
Best time to relax           17.647059
Friends are available then    11.764706
```

We now see that most people are motivated to watch at their most commonly viewed time by the fact that the games are played then, or that it fits their schedule well. Game time being the leading factor is not shocking and truly dictates when you are tuning in.

Why do you usually watch sports in that location?

17 responses



why_location	
Most comfortable	41.176471
Social environment	29.411765
Easy access	23.529412
Best viewing experience	5.882353

We see here that most people are most motivated to watch at their most common location, either being the most comfortable or for the social environment. I would agree with this distribution as I tend to associate watching sports with being around friends, or if by myself, whichever place is most convenient is typically my choice. Finally, I think this would be true for a large section of the population as well.

How:

The survey was created using Google Forms and was sent out to Berry College students. Approximately 2 weeks were given to complete the survey, analysis, and report, whereas the responses received were voluntary and anonymously collected over the course of a week. After receiving 17 responses, the responses were downloaded as a csv directly from Google Forms. Before starting the actual analysis, the csv was cleaned and organized into a format easier for analysis, such as removing the unnecessary time stamps and changing the column names to be easier to manage. Next, the data analysis was conducted in Python utilizing the pandas library. Pandas data frames were used to calculate percentages for the questions through the `value_counts` method to determine the distribution of responses related to each question. Several limitations include the small sample size and the lack of random selection. Furthermore, the demographic makeup of the participants was very similar due to many of the respondents being friends and teammates of mine. This lack of variation could be improved upon by trying to survey more broadly rather than just small subsets. Finally, the study could be improved by increasing the number of responses, getting a wider sample, and asking more open-ended questions in order to get more honest responses rather than generic ideas from multiple-choice.

Discussion/Conclusion:

Overall, as it stands for picking up on any patterns or trends, I do not believe any strong or credible trends were discovered. However, the lack of variation in the sample may be causing the lack of trends, especially considering how small the sample is. On the contrary, the only ‘trend’ I somewhat see is that most responses usually have two pretty dominant responses, and the other responses are way lower proportions. In the end, not really any meaningful trends stick out.